

Week 2

Aim

The aim of this seminar was to understand more about key encryptions, specifically RSA encryption and how to verify the identity of the sender and trust the integrity of the message which is being received. This seminar specifically utilises the cryptography library to generate keys and learn more about their usage.

Glossary

Public Key – Asymmetric encryption that can be shared openly.

Private key - Secret key in asymmetric cryptography

Asymmetric – Uses two keys, one for decryption, one for encryption.

Symmetric – Uses one key for both encryption and decryption

Digital signature – Cryptography method to prove integrity and authenticity of message

RSA algorithm - Popular asymmetric encryption algorithm based on factoring large prime numbers

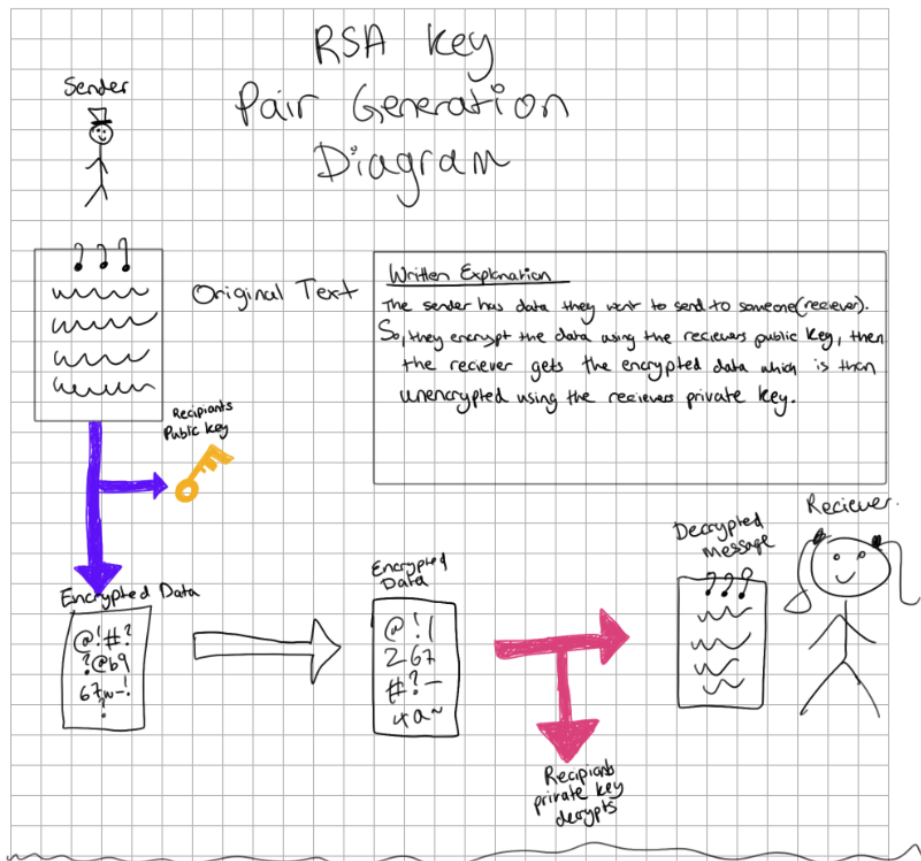
AES algorithm – Uses the same key for both encrypting and decrypting unlike RSA

Hash - One-way deterministic function that transforms output into a fixed size output.

Code/Description

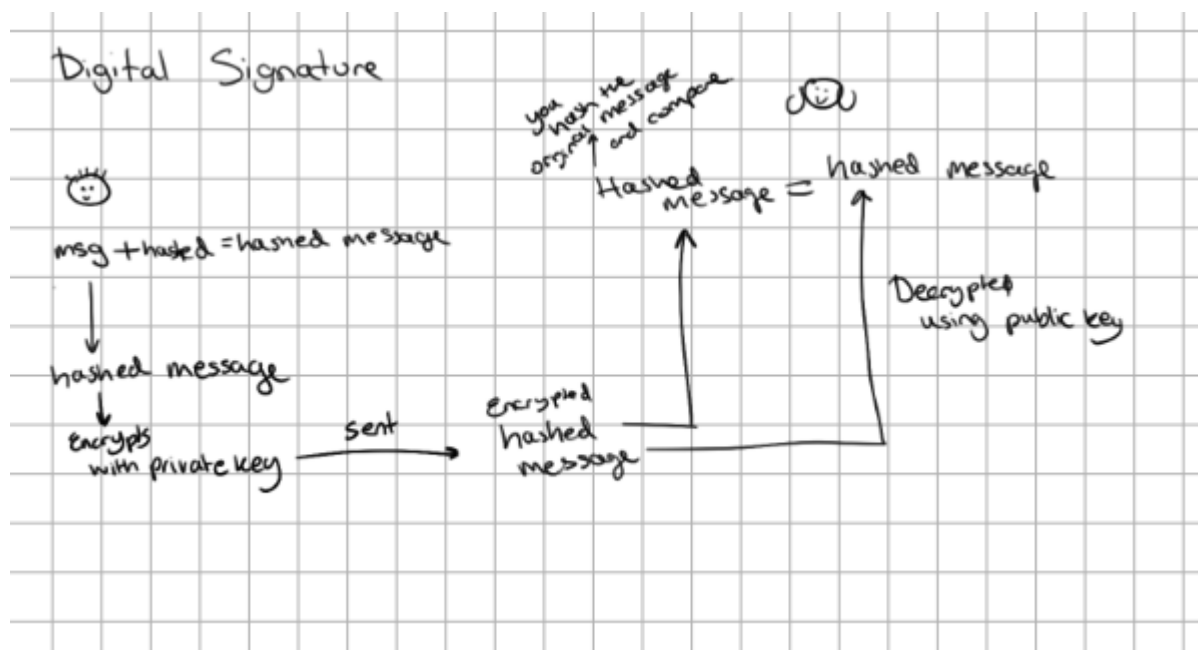
RSA algorithm

Within this seminar, we covered RSA key pair generation and how key pairs are used to encrypt and decrypt messages. Throughout this seminar, I attempted to comment my code thoroughly to explain the process step by step. I have also created diagrams explaining key concepts as you see below.



The RSA key pair are two mathematically linked keys which consist of a public and private key. This is used in asymmetric encryption where the public key encrypt data or verifies signature and the private key decrypts data or creates the signature.

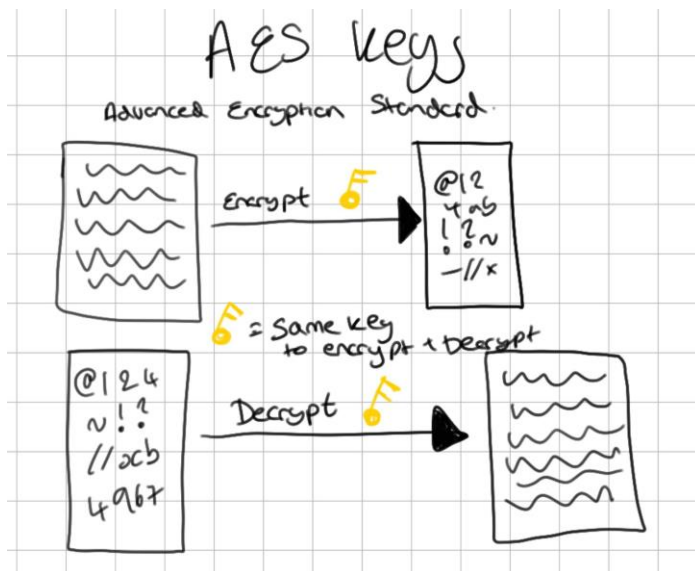
Digital Signature



A digital signature is a method used to verify the identity of the sender for the receiver, it is similar to a fingerprint and it utilises a user's private key which is only known to them to encrypt a hash of the

data, which the receivers then uses the senders public key to verify that the data has not been changed seeing how if the value has been changed then the comparison of the final value would be different and therefore tampered with.

Another main aspect to help understand this concept more is that the sender would never use their own public key to encrypt data. Therefore, when a message is sent, the sender uses either the receivers public key or their own private key to encrypt. Otherwise, if they use their own public key, nobody will have access to their private key to decrypt the data.



Issues

One main issue with this topic that I struggled to understand was digital signatures. This was because there were many steps to this process which made it hard to follow without mixing up the keys. Alongside with this, understanding what is being encrypted versus hashed was also hard to follow.

In order to overcome this issue, I decided to use websites that allowed for a more interactive approach where I would generate a hash of a secret message, that I would encrypt this hash with a secret key and then decrypt with my secret key and compare the final hashes value. Although it was a more symmetric approach, it helped me to understand that towards the end of the transaction, the final hash values would be compared.